



Assignment Title	Design and Implementation of a Data Warehouse and Business Intelligence Solution
Learning outcomes covered	LO1 – LO4
Assignment Mode	Take-home
Maximum Marks	200
Contribution to the Final Grade	20%
Date published	
Deadline for submissions	
Mode of Submission	Report + Technical Files

Assignment Description

Students are required to design and implement an end-to-end Data Warehouse and Business Intelligence solution using a selected real-world business scenario. The activity requires students to collect and prepare operational data sources, integrate data using ETL processes, design a dimensional data warehouse, develop analytical solutions, and present meaningful business insights through BI dashboards.

Students must demonstrate practical understanding of concepts covered throughout the module, including data warehouse architecture, dimensional modelling, ETL processes, OLAP analysis, and business intelligence reporting.

Students should document all implementation steps followed during the activity. The report should include relevant diagrams, screenshots, explanations, and analytical findings.

Task 1: Dataset Selection and Business Scenario Identification (10 Marks)

Students must select a suitable real-world dataset representing an operational business environment.

The selected dataset should:

- Represent an OLTP system.
- Contain sufficient records for analytical processing.
- Include multiple attributes suitable for dimensional analysis.
- Support the creation of fact and dimension tables.
- Represent a realistic business scenario.

Suggested domains:

- Retail / E-commerce, Healthcare, Banking and Finance, Tourism, Education, Transportation, Social Media, IoT and Smart Cities

Students must document:

Dataset Overview

- Business domain
- Purpose of the system
- Business problem addressed
- Dataset source
- Dataset format
- Number of records
- Attributes description

Students should explain why the selected dataset is suitable for developing a Data Warehouse and BI solution.

Task 2: Data Source Identification and Preparation (10 Marks)

Identify and prepare the data sources required for the solution. Students should use multiple source types where possible.

Examples:

- CSV files
- Excel files
- Relational databases
- JSON files
- APIs

Documentation should include:

- Description of each data source
- Source format
- Available attributes
- Relationship between sources
- Data preparation steps

Students should explain how data from different operational systems can be integrated into a unified analytical environment.

Task 3: Data Warehouse Architecture Design (10 Marks)

Design a high-level Data Warehouse and Business Intelligence architecture.

The architecture should include:

Data Source Layer

Examples:

- Operational databases
- External files
- Application systems

Data Integration Layer

Include:

- Extraction
- Transformation
- Loading process

Storage Layer

Include:

- Staging area
- Enterprise Data Warehouse
- Data Mart

Presentation Layer

Include:

- OLAP
- Reporting tools
- BI dashboards

Students must provide:

- Architecture diagram
- Explanation of each component

Task 4: Dimensional Data Warehouse Design and Implementation (20 Marks)

Design and implement a dimensional data warehouse. Students should apply dimensional modelling principles. The solution should include:

Fact Table

Define:

- Business process
- Grain
- Measures
- Foreign keys

Dimension Tables

Include appropriate dimensions such as:

- Customer Dimension
- Product Dimension
- Location Dimension
- Date Dimension

Students should implement:

- Star Schema or Snowflake Schema
- Primary keys and foreign keys
- Relationships and Hierarchies

Documentation should include:

- Schema diagram
- Table descriptions
- Design assumptions

Task 5: ETL Process Development (20 Marks)

Develop a complete ETL pipeline.

Extract

Extract data from selected sources.

Document:

- Extraction method
- Source connection
- Data extraction process

Transform

Perform required data transformations:

Examples:

- Missing value handling
- Duplicate removal
- Data cleansing
- Standardisation

- Data type conversion
- Aggregation
- Derived attribute creation

Load

Load transformed data into:

- Dimension tables
- Fact tables

Students should document:

- ETL workflow
- Transformation logic
- Loading sequence
- Validation results

Task 6: Data Mart Development (10 Marks)

Create at least one department-specific Data Mart.

Students must explain:

- Purpose of the data mart
- Target users
- Analytical benefits

Task 7: OLAP Analysis and Business Intelligence Dashboard Development (15 Marks)

Develop interactive dashboards using:

- Microsoft Power BI or Tableau

The dashboard should contain:

Report Page 1: Executive Summary

Include:

- KPIs
- Summary cards
- Overall performance indicators

Report Page 2: Trend Analysis

Include:

- Time-based analysis
- Line charts
- Comparison charts

Report Page 3: Interactive Analysis

Include:

- Filters, Slicers, Drill-down, Roll-up

Task 8: Business Insights and Recommendations (5 Marks)

Analyse dashboard outputs and provide meaningful insights. Minimum five insights should be presented.

Examples:

- Sales trends
- Customer patterns
- Operational performance
- Regional comparison
- Future recommendations

Students should explain how the insights support business decision-making.

Usage of AI: The use of AI tools is allowed for this assessment as a supporting tool for learning and development. Students may use AI tools to assist with brainstorming, improving explanations, generating ideas, debugging, and refining their work. However, students must ensure that the final submission reflects their own understanding and implementation. AI-generated content should be critically evaluated, verified, and appropriately adapted. Students must not use AI to fully generate their assignment, create false results, or replace their own technical work. Any use of AI tools should be acknowledged in the report with a brief explanation of how AI was used. This assessment follows the CLEAR framework, encouraging responsible and transparent use of AI.

Marking Scheme

Criteria	Excellent	Good	Satisfactory	Poor	Very Poor
Criterion 1 – Business Scenario, Dataset & Data Sources (Tasks 1–2)	Highly suitable realistic OLTP dataset; business problem, source, attributes and relationships are comprehensive and well justified. Multiple sources are appropriately prepared and integration is clearly explained.	Suitable dataset and business scenario with good documentation. Sources and preparation are mostly complete with minor omissions.	Acceptable dataset and basic scenario. Sources are identified but justification, preparation or relationships lack depth.	Weak dataset/scenario selection with limited source description and preparation. Important requirements are missing.	Unsuitable or inadequately documented dataset; little/no meaningful source identification or preparation.
Criterion 2 – DW Architecture & Dimensional Modelling (Tasks 3–4)	Excellent end-to-end architecture with clearly justified layers. Dimensional model has appropriate grain, fact/measures, dimensions, keys, hierarchies and relationships. Schema is technically correct and professionally documented.	Sound architecture and dimensional model with appropriate components and mostly correct relationships; minor design/documentation issues.	Functional basic architecture and schema, but some modelling decisions, relationships, grain or documentation lack clarity.	Major weaknesses in architecture or dimensional design; incomplete schema, incorrect relationships or insufficient justification.	Architecture/model is absent, largely incorrect or does not demonstrate dimensional modelling principles.
Criterion 3 – ETL Implementation & Data Mart	Complete, robust and reproducible ETL pipeline covering extraction,	ETL pipeline and Data Mart operate correctly with good	Basic working ETL and Data Mart with limited transformation	Partially functioning ETL/Data Mart; significant	ETL/Data Mart is absent, non-functional or demonstrates

(Tasks 5–6)	meaningful transformations, loading sequence and validation. Data Mart is correctly implemented and strongly aligned to users and analytical requirements. Clear process evidence is provided.	transformations and documentation; minor omissions in validation or process evidence.	complexity, validation, documentation or justification.	errors, weak transformations or inadequate evidence of successful loading.	little understanding of the required process.
Criterion 4 – OLAP & BI Dashboard (Task 7)	Professional, interactive BI solution with meaningful KPIs, executive summary, trend analysis, suitable visualisations, filters/slicers and effective drill-down/roll-up. Strong usability and analytical value.	Well-designed functional dashboard containing most required analytical and interactive features; minor usability/analytical issues.	Functional dashboard with basic KPIs, charts and interaction but limited analytical depth or design quality.	Dashboard is incomplete or weakly designed with limited functionality, inappropriate visuals or missing interactive features.	Dashboard is absent, largely non-functional or does not satisfy the BI/OLAP requirements.
Criterion 5 – Business Insights & Recommendations (Task 8)	Five or more clear, evidence-based insights derived directly from dashboard results. Recommendations are specific, practical and strongly linked to business decision-making.	Good insights supported by dashboard evidence with relevant recommendations and clear business relevance.	Required insights are provided but analysis or recommendations are relatively descriptive/basic.	Limited or weakly supported insights; recommendations lack relevance or connection to dashboard findings.	Insufficient, unsupported or incorrect insights and recommendations.